

Who Pays More, and Who Can Least Afford It? Food Price Premiums, Deprivation and Abnormal Price Movements Across Selected Pakistani Urban Markets

Milestone 02: data preparation, exploratory analysis and hypotheses (draft Sections II and III)

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Code and data: github.com/Samzzh0803/Pakistan-Food-Price-Premiums-and-Deprivation (branch milestone-02)

Summary. Pakistan's weekly Sensitive Price Indicator (SPI) reports one national figure. Beneath it, PBS publishes prices for 51 items in 17 cities. We parse all 27 weekly releases available up to our freeze date (2025-05-15 to 2026-09-24; 23,409 city-item-week rows). The parse reproduces PBS's own national minimum, maximum and geometric-mean average in every one of 1,377 item-weeks. We merge in district deprivation (PSLM 2019-20), population (Census 2023) and urban food CPI. On 32 food items, Islamabad pays +14.1% and Rawalpindi +10.2% above the national reference, while Bannu (-7.4%) and Sukkur (-7.1%) pay less. 4 of 17 city premiums survive a Bonferroni correction. Premiums are persistent: 93% of above-median city-item pairs stay above the median the next week. However, 75% of weekly food price changes are exactly zero. Across the 17 cities, the premium is *negatively* rank-correlated with district deprivation (Spearman $\rho = -0.56$, $p = 0.020$): the cities paying most are mostly not the ones least able to absorb it. How many separate quotes stand behind a city's average varies widely, which matters for every later model.

Changes since Milestone 01

Milestone 01 reported a 7-city panel. That was a parsing error. Appendix-A of the PBS Annexure is three tables stacked in one sheet, and the earlier script stopped after the first. The panel actually covers all 17 SPI cities (8 Punjab, 4 Sindh, 2 Khyber Pakhtunkhwa, 2 Balochistan, plus Islamabad), so the "Punjab and Islamabad only" limitation is retired. The earlier figure of 16,380 rows, explained then as records times MIN/AVG/MAX, was a coincidence of reading three blocks under one set of city names. The faulty files are kept as an audit trail and a dated correction note is in the repository (reports/CORRECTION_2026-10-04_appendix_a_17_cities.md). Following the instructor's feedback that a price-only study is too narrow, we now merge three external sources, each with one job (Table 1). The reference price is the geometric mean across quoting cities, which is PBS's own definition (§2.1). Hypothesis H5 (burden versus price) is new and is now testable with $n = 17$.

1. Alignment and inventory

Research question. Beneath the single national weekly SPI figure, do persistent city-level food price differences exist, are the cities paying most the ones least able to absorb it, and is the published city detail enough to detect localised price stress? **SDGs:** primary SDG 2 (Target 2.1 access to food; Target 2.c functioning food markets and timely market information), secondary SDG 10 (reduced inequalities). **Division of perspectives:** both members are computer scientists, so we split roles. One owns acquisition, parsing and provenance. The other owns statistical framing, representation, ethics and policy interpretation.

Table 1. Data inventory. Observational unit: one city x item x week. Master dataset: 23,409 rows x 46 columns; 17 cities, 51 items (32 food), 27 weeks.

Source	Role	Vintage / coverage	Access	Size used
PBS weekly SPI Annexure, Appendix-A	Prices: MIN/AVG/MAX by city and item (the panel)	27 weeks, 2025-05-15 to 2026-09-24	xlsx from pbs.gov.pk release pages	23,409 rows (867/week)
PSLM District Level Survey 2019-20	Deprivation (burden denominator): FIES, literacy, schooling, water	2019-20	PDF report tables, text extraction	22 districts
Pakistan Digital Census 2023, Table 1	Population: total and urban, for size and fairness strata	2023	xlsx by province	23 districts
CPI (Urban) Food & non-alcoholic beverages	Deflator only (not a regressor)	monthly 2025-05 to 2026-09	PDF + Monthly Review docx	13 panel months

2. Ingestion and preparation

2.1 Collection audit and validation

Discovery. PBS publishes each weekly release on a page of the form /weekly-sensitive-price-indicator-spi-for-the-week-ended-on-DD-MM-YYYY/. That page links the Annexure workbook. We swept 73 Thursdays from 2025-05-15 onward, trying the Thursday, Wednesday and Friday page for each (197 requests, rate-limited, every attempt logged in data/raw/pbs_backfill_manifest.csv with URL, SHA-256 and time). Release pages exist for only 12 of those weeks. Direct file URLs for older weeks return 404 under every naming variant we tried, and site search lists only recent releases. The panel therefore holds 27 weeks: 27 distinct workbooks, after dropping 12 byte-identical duplicates by hash, keeping the earlier repository copy each time. Every re-downloaded file was byte-identical to its earlier copy, which independently confirms provenance. **We do not describe the archive as complete.** Earlier routes that failed or were rejected are documented in the repository audits: the Punjab AMIS wholesale portal (HTTPS fails TLS negotiation; HTTP pages load but historical queries were not reproducible and no commodity or market match could be verified), an older single-pattern URL sweep that tried Thursdays only, and PDF annexures (superseded by xlsx). The week of 2026-10-01 falls after the freeze (2026-09-24). It was saved to an append-only holdout folder, hashed and never parsed.

Validation. The parser finds every row whose second cell is DESCRIPTION and reads the city labels Name (NN) on the row above. Columns under any other header are not cities. It reads the week from the PRICES ON dd-mm-yyyy cell and checks it against the file name. Assertions fail loudly on every week: 3 blocks, 17 cities, 51 items, 867 rows, no duplicate key, $\text{MIN} \leq \text{AVG} \leq \text{MAX}$ where positive, and zeros only as all-zero triples. Against PBS's own national columns, the cross-city minimum and maximum equal PBS's National MIN and MAX in 1,377 of 1,377 item-weeks. The **geometric** mean of the quoting cities' averages matches PBS's National AVG within 0.1% in 1,377 of 1,377 (worst case 0.046%). The arithmetic mean misses by a median of 0.42% and the median by 1.07%. The geometric mean is therefore PBS's reference price and ours. Release dates are 25 Thursdays and 2 Wednesdays. There are 15 consecutive pairs (14 at 7 days, 1 at 6). Other gaps run from 13 to 77 days (Fig. 1).

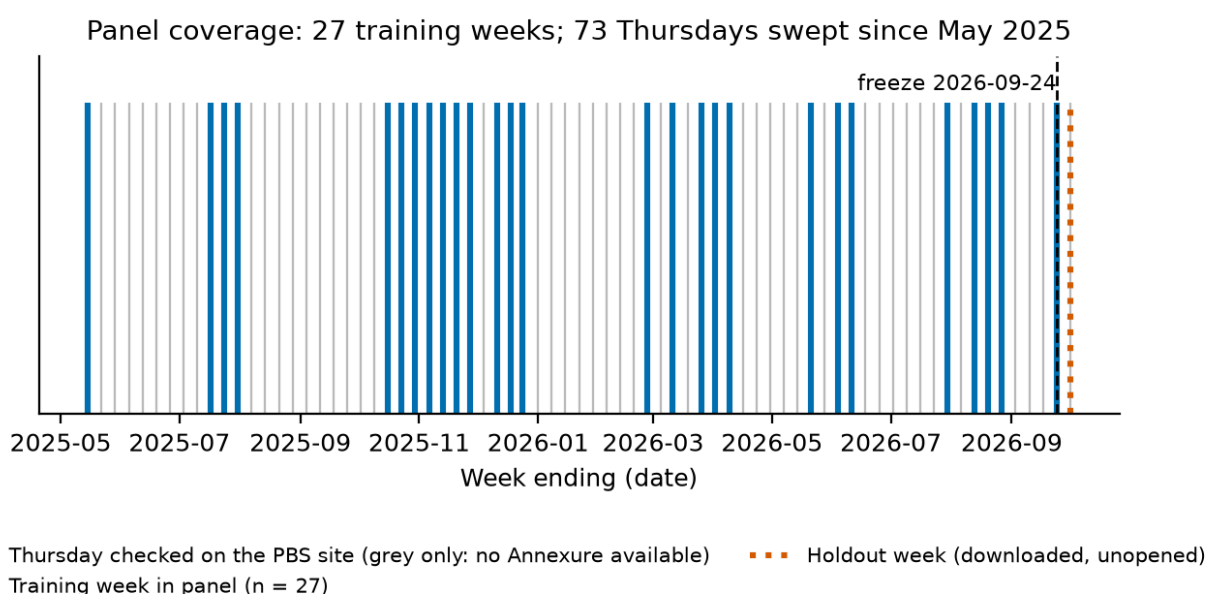


Fig. 1. Weeks in the training panel (blue), Thursdays checked on the PBS site (grey) and the frozen holdout week (orange). Week changes are only computed across the consecutive pairs; the gaps are a coverage limitation, not filled.

2.2 Reshaping and cleaning (Unit 03)

The raw sheet (179 rows x 25 columns) breaks all three tidy-data rules. The variable city is stored in headers. One observation is spread over MIN, AVG and MAX cells under a merged city header. Three tables share one sheet. We melt it to one row per city, item and week (867 rows per week), keeping PBS's serial as `item_id` and its city code as `city_code`, both nominal. The national summary columns of block 3 go to a separate reference table and are never panel rows. Table 2 lists every cleaning action. Nothing is deleted.

Table 2. Cleaning log (counts written by the pipeline).

operation	rows_or_files	note
City labels with hyphen or line break repaired	0	Repair rule in parser; none needed in the 27 training files
Zero MIN/AVG/MAX triples set to missing	83	Structural non-quotes; PBS excludes them from its national mean
Week dates read from 'PRICES ON' text	27	All match the date in the file name
Item label wording changes mapped to first label	17	Item 42 renamed in 2026-09-24 release, unit and price unchanged
Duplicate workbooks dropped (identical sha256)	12	Earlier repo copy kept; re-downloads byte-identical
Rows deleted	0	Nothing deleted; missing values are flagged, not dropped

2.3 Missing data

Of the options taught (remove, impute from internal data, impute from external data, constant), none is applied to prices. There are 83 structural non-quotes (0.35% of rows), where PBS prints 0/0/0: rice IRRI-6/9 in Gujranwala, Lahore and Sialkot in every week, and gas charges in Khuzdar in two weeks. PBS itself leaves these cities out of its national mean, so imputing would invent a market price the source does not record. Weekly changes across non-consecutive gaps (10,367 rows, 44.3%) stay missing, because interpolation would manufacture changes nobody observed. Missing weeks are a coverage gap. The trade-off is statistical convenience against selection bias: complete-case analysis is honest here only because the missingness is structural and documented (Table 3; per-column counts are in the notebook and codebook).

Table 3. The three kinds of missingness and the choice made for each.

kind	rows	choice	reason
Structural non-quote (PBS printed 0/0/0)	83.000	set to missing, flag, do not impute	Imputing invents a market price PBS does not record; PBS itself excludes these cities from its national mean
Weekly change across a non-consecutive gap	10,367.000	leave missing	Interpolation would manufacture week-to-week changes nobody observed
Weeks absent from the PBS site	–	treat as a coverage gap	No source to impute from; stated as a limitation, not filled

2.4 Transformations, scaling, encoding and discretisation (Unit 04)

Log price. Prices behave multiplicatively, so a 10% premium means the same thing on salt and on a 20 kg flour bag. Pooled raw PKR across items is not a meaningful distribution: it mixes units, with skewness 2.13. Within an item, the median skewness falls from 0.216 on raw prices to 0.024 on log prices. All cross-city comparisons use the relative price $rel_price = \log(\text{price} / \text{national geometric mean})$. By construction it sums to zero across quoting cities in each item-week, which is exactly the sum-to-zero premium in H1. **Redundancy:** log MIN and log AVG correlate at 0.998, so we keep AVG plus the within-city range. Provenance columns are kept for audit and excluded from analysis. **Normalisation:** min-max scaling (0 to 1, higher = more deprived) for the deprivation indicators; z-scores of price within item and week. Decimal scaling divides by a power of ten that differs by item (10^4 for flour), so it does not make items comparable. Scaling matters because the distance-based methods in §3.7 would otherwise be dominated by large-valued features. All constants come from the training panel. **Encoding:** 0/1 dummies for province and item category, and sum-to-zero (deviation) coding for cities, so each city coefficient is a premium relative to the average city. **Splitting:** `week_end` is split into year, month and ISO week (13 calendar months). **Discretisation:** equal-width bins collapse on population because of Karachi (Table 4), so the strata used for H4 and the contingency tables are equal-frequency terciles.

Table 4. Equal-width against equal-frequency binning (cities per bin).

variable (17 cities)	method	low	mid	high
pop_urban	equal-width	15	0	2
pop_urban	equal-frequency (terciles)	6	5	6
deprivation_composite	equal-width	9	6	2
deprivation_composite	equal-frequency (terciles)	6	5	6

2.5 Integration and the external sources

Stacking joins the three blocks and the 27 weeks. Merging adds the item classification (on `item_id`), the city attributes (on `city`) and the CPI (on the calendar month of `week_end`). Each merge is many-to-one, and the row count is asserted unchanged (23,409; Table 5). **Crosswalk decisions** (Appendix Table A3): Karachi is one SPI market but seven Census 2023 districts and six PSLM 2019-20 districts. Keamari was split from Karachi West after the survey, so PSLM values are population-weighted with Keamari added to West's weight. Islamabad and Rawalpindi form one contiguous market but are kept as two administrative units, each with its own district values; §3.7 shows they are each other's nearest neighbour in price space. PSLM has no urban sample for Khuzdar, and FIES is published only for district totals, so all four indicators use **district totals** for comparability. The urban values are kept as a sensitivity column. The four deprivation indicators, all oriented so that higher means more deprived, are moderate-or-severe food insecurity (FIES, SDG 2.1.2; 7.9% in Islamabad to 29.3% in Khuzdar), illiteracy at age 10+, children 5-16 out of school, and households without tap water. They are min-max scaled and averaged into a transparent composite. The water indicator partly reflects infrastructure type (much of urban Punjab uses motor pumps), so a composite without it is reported as a sensitivity check. These are **district deprivation measures from 2019-20, not city income**: PBS publishes income only by province. **CPI**: urban food CPI rose 14.0% from 2025-05 to 2026-09. It is used only to express real price levels. H1, H2 and H5 compare cities within the same week, so no national deflator can change them, and a nationally uniform series would be collinear with week effects as a regressor.

Table 5. Stacking and merging steps, keys, cardinality and row counts.

operation	source	key	cardinality	rows_after
stack	3 Appendix-A blocks per workbook	same 51 item rows	rows: 7+7+3 cities x 51 = 867 per week	867
stack	27 weekly workbooks	week_end	867 x 27	23409
merge	item_mapping.csv	item_id	many-to-one (51 items)	23409
merge	city table (Census 2023 + PSLM 2019-20)	city	many-to-one (17 cities)	23409
merge	urban food CPI	calendar month of week_end	many-to-one (months)	23409

3. Exploratory data analysis

Table 6 summarises the key variables at their own unit of analysis. City attributes have 17 distinct values and item comparisons about 32, so every city-level or item-level test below is run on aggregated values, never on the 23,409 rows.

Table 6. Summary statistics (mean, median, SD, IQR) for key variables. `rel_price` and `dlog` are in log points.

variable	n	mean	median	sd	iqr	min	max
rel_price (food, log points)	14607	-0.000	0.000	0.130	0.110	-1.239	0.708
dlog_price (food, consecutive weeks)	8115	-0.003	0.000	0.072	0.000	-0.981	0.709
within_city_range_pct (food)	14607	6.812	1.395	11.665	9.384	0.000	152.858
single_quote (food, share)	14607	0.486	0.000	0.500	1.000	0.000	1.000
real_price_avg / price_avg (deflator ratio)	14607	0.342	0.341	0.012	0.013	0.321	0.366
deprivation_composite (17 cities)	17	0.427	0.349	0.217	0.261	0.138	0.974
fies_mod_sev_pct (17 cities)	17	15.314	16.380	5.926	8.310	7.930	29.260
pop_urban, millions (17 cities)	17	3.524	1.619	4.940	2.485	0.048	18.868

3.1 City premiums (H1)

For each city we average `rel_price` over weeks for each food item. The observations are then those item-level means, so `n` is the number of food items: 31 for the three cities with structural rice gaps, 32 otherwise. Islamabad's mean premium is +14.1% (95% t-interval [8.9, 19.6]), Rawalpindi +10.2% and Karachi +5.2%. At the other end are Sukkur (-7.1%) and Bannu (-7.4%) (Fig. 2, Table 7). In two-tailed tests of H_0 : mean premium = 0, 8 of 17 cities differ at 0.05. Seventeen tests inflate the Type I error, so at the Bonferroni threshold $0.05/17 = 0.0029$ only 4 survive: Islamabad, Rawalpindi, Sukkur, Bannu. A Type I error here would tell a policymaker that a city's consumers pay more when they do not. A Type II error would miss a real premium, a cost borne by that city's households. Resistant and non-resistant summaries diverge: Larkana's mean premium (-4.9%) and median (-0.2%) differ by 4.8 points, because a few items pull the mean. We report both and treat the median as the safer summary where they disagree. Bootstrapping items (2000 resamples) puts intervals that exclude zero on the median premium of 6 cities. The 17 cities are PBS's purposive selection, so these intervals describe the item basket within these markets, not Pakistan's urban areas at large. An exploratory regression of log price on item-week effects and sum-to-zero city effects ($n = 14,607$ rows) returns city coefficients identical to the simple means ($r = 1.000$), as expected in a near-balanced panel. Its F statistic for H1a ($F = 179.3$, $df = 16, 13,727$) treats rows as independent, which repeated weekly prices are not, so we read it descriptively.

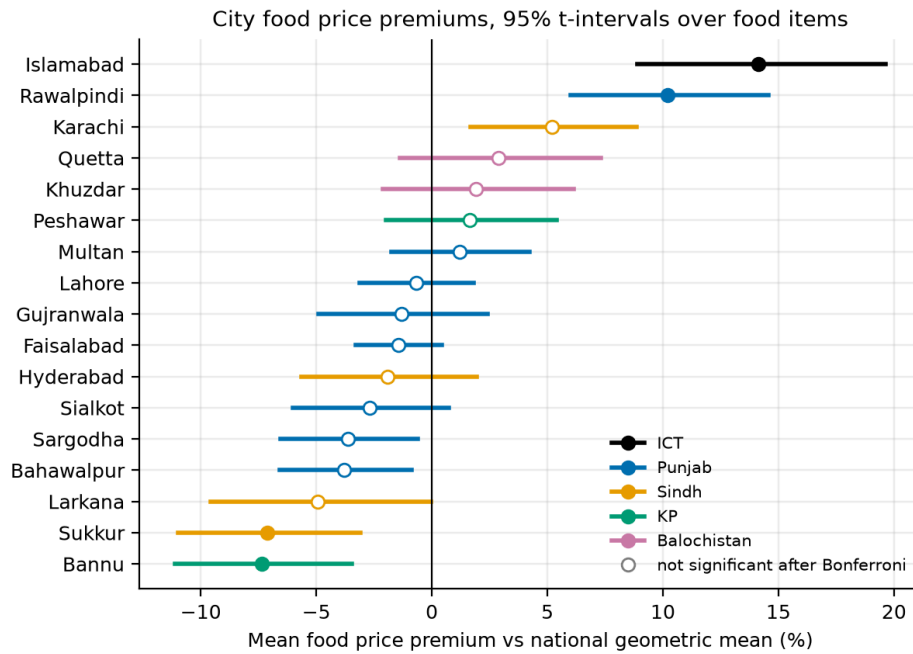


Fig. 2. Mean food premium per city with 95% t-intervals (n = food items). Filled markers survive Bonferroni.

Table 7. City food price premiums against the national geometric mean (percent = $\exp(\text{mean log premium}) - 1$).

city	province	n items	mean %	95% CI (%)	median %	t	p	Bonferroni
Islamabad	ICT	32	14.14	[8.9, 19.6]	12.64	5.73	< 0.001	yes
Rawalpindi	Punjab	32	10.21	[6.0, 14.6]	6.79	5.10	< 0.001	yes
Karachi	Sindh	32	5.22	[1.7, 8.9]	2.63	3.04	0.005	no
Quetta	Balochistan	32	2.87	[-1.4, 7.3]	3.16	1.37	0.181	no
Khuzdar	Balochistan	32	1.92	[-2.1, 6.1]	2.32	0.96	0.345	no
Peshawar	KP	32	1.65	[-2.0, 5.4]	0.41	0.92	0.365	no
Multan	Punjab	32	1.19	[-1.8, 4.2]	0.69	0.82	0.420	no
Lahore	Punjab	31	-0.68	[-3.1, 1.8]	0.04	-0.56	0.580	no
Gujranwala	Punjab	31	-1.31	[-4.9, 2.4]	-0.72	-0.73	0.473	no
Faisalabad	Punjab	32	-1.45	[-3.3, 0.4]	-0.14	-1.56	0.128	no
Hyderabad	Sindh	32	-1.93	[-5.6, 1.9]	-0.10	-1.03	0.313	no
Sialkot	Punjab	31	-2.69	[-6.0, 0.7]	-0.20	-1.61	0.119	no
Sargodha	Punjab	32	-3.62	[-6.5, -0.6]	-2.63	-2.45	0.020	no
Bahawalpur	Punjab	32	-3.78	[-6.6, -0.9]	-2.45	-2.65	0.013	no
Larkana	Sindh	32	-4.92	[-9.6, -0.0]	-0.16	-2.06	0.048	no
Sukkur	Sindh	32	-7.11	[-11.0, -3.1]	-4.05	-3.54	0.001	yes
Bannu	KP	32	-7.36	[-11.1, -3.5]	-4.57	-3.77	< 0.001	yes

3.2 Dispersion by item type (H2)

The coefficient of variation compares spread across items priced in different units. Mean cross-city CV is highest for perishables (0.157, 11 items) and prepared food (0.144), lower for storable staples (0.107, 10 items), and lowest for branded packaged goods (0.026), which carry printed or distributor prices (Fig. 3). A one-tailed comparison of perishables against storables gives $SE(A - B) = 0.0256$, Welch $t = 1.92$, $p = 0.037$. The rank-based check gives $p = 0.057$. With about ten items a side this is marginal evidence, and we do not oversell it. Items were classified on storability and pricing mechanism before looking at dispersion.

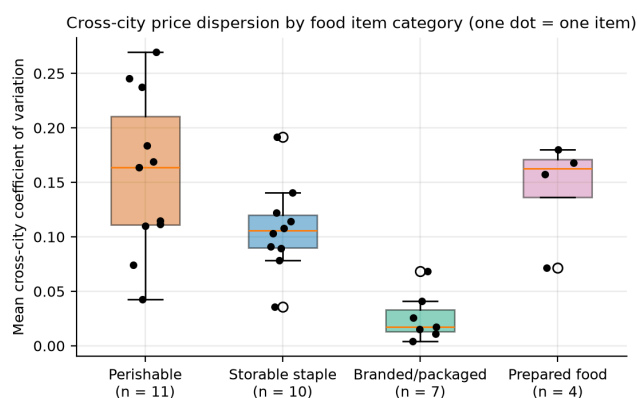


Fig. 3. Mean cross-city CV per food item, by category. Branded goods barely vary across cities; perishables vary most.

3.3 Persistence and sticky prices (H1b, H3)

Across 3,269 consecutive-week pairs where a city was above the cross-city median, 92.7% stayed above it the next week. City rates range from 82% to 97%, and a one-tailed test of the 17 city rates against 0.5 rejects decisively ($t = 37.0$, $p < 0.001$). Persistence is partly mechanical, however: 75.1% of weekly food log changes are exactly zero (56% for perishables, 75% for storables, 92% for branded and 98% for prepared items). Restricted to pairs where the price changed (916 pairs), the stay-above rate is still 84.1% ($t = 7.3$, $p < 0.001$), with the lowest city at 52%. That stickiness reshapes H3. Under the contracted rule (abnormal = |change| above the item's own 90th percentile), the threshold is zero for 13 of 32 food items, so any movement at all would count as abnormal (§4.2).

3.4 Distributions and outliers

Relative prices are roughly symmetric (skew -0.14) but heavy-tailed (excess kurtosis 3.47), with a spike at zero from items priced identically everywhere (Fig. 5). Across all 51 items, the quartile method flags 3,649 of 23,326 values as mild (beyond 1.5 IQR) and 1,251 as regular (beyond 3 IQR). The mean-and-SD method flags 1,598 beyond 2 SD and 434 beyond 3 SD. The methods disagree because the narrow IQR, set by the many zero-premium items, puts the fences close, while the SD is inflated by the very extremes it is meant to find, which masks them. On weekly changes the IQR is exactly 0, so both quartile tiers flag every non-zero change (2,502). That is a degenerate result and the method is unusable there. We checked the eight largest |rel_price| values against the raw workbooks: 7 equal PBS's own national MIN or MAX for that item-week. The other (Sukkur, Tomatoes, week 2025-05-15) has MIN = 20, AVG = 20, MAX = 20 PKR in the raw sheet, a genuine quoted price. All are kept and flagged, because large moves are what H3 studies. The CLT supports t-intervals on city means over about 32 items, but the item-level means are skewed for some cities (median skewness across cities -0.29), which is one more reason to report medians and bootstrap intervals alongside.

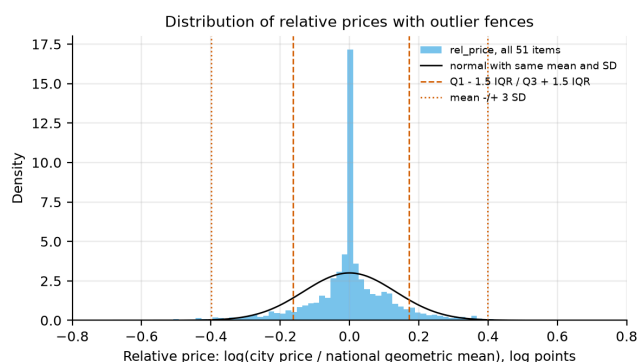


Fig. 5. Relative prices with the 1.5 IQR and 3 SD fences and a fitted normal curve. Heavy tails and a spike at 0.

3.5 Who pays more versus who can least afford it (H5)

Following the Anscombe lesson, we plot before computing (Fig. 7). The relationship runs **against** the "poor pay more" expectation. Premiums are highest in the least deprived cities (Islamabad, Rawalpindi, Karachi), and several of the most deprived (Bannu, Sukkur, Larkana) pay below the national reference. Spearman ρ between premium and the deprivation composite is -0.56 ($p = 0.020$, $n = 17$). Dropping each city in turn keeps ρ between -0.72 and -0.48, and the most influential city is Khuzdar. With FIES alone, $\rho = -0.32$ ($p = 0.209$), not significant. Without the water indicator, $\rho = -0.50$ ($p = 0.041$). A simple line of premium on the composite has slope -0.127 log points per unit of deprivation (95% CI [-0.249, -0.004], $R^2 = 0.24$). This is an association across 17 cities, not a causal effect, and the deprivation data predate the prices by six years. The burden re-ranking (Fig. 8) shows the policy point. The five highest-premium cities (Islamabad, Rawalpindi, Karachi, Quetta, Khuzdar) share 1 member with the five most deprived (Khuzdar, Larkana, Bahawalpur, Sukkur, Bannu) and 2 with the five most food-insecure. Combining premium with need, the top three cities for burden are Khuzdar, Quetta, Multan under the composite definition and Khuzdar, Quetta, Bahawalpur under the FIES definition. Under both, burden concentrates in Khuzdar and Quetta, whose premiums are above the reference (though not individually significant) and whose food insecurity is the highest in the panel.

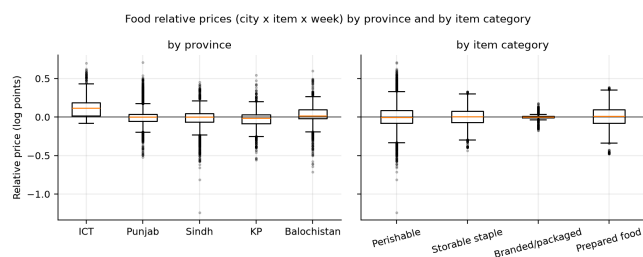


Fig. 4. Food relative prices by province and by item category (city x item x week).

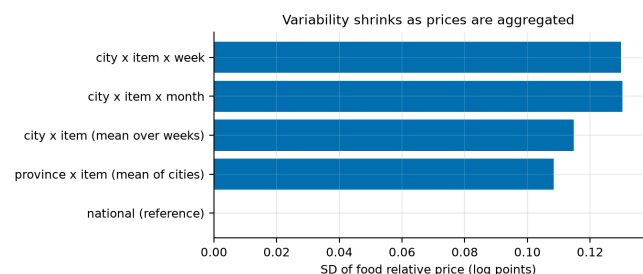


Fig. 6. SD of food relative prices at successive levels of aggregation; the national headline sets it to 0.

City food price premium against district deprivation (PSLM 2019-20)

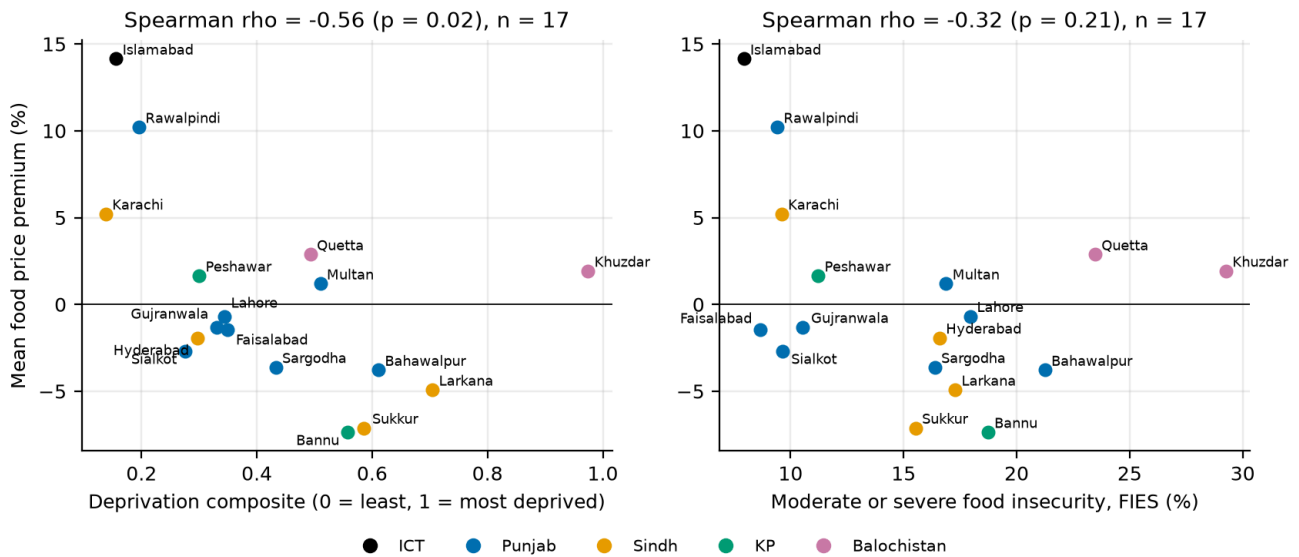


Fig. 7. City premium against the deprivation composite (left) and FIES food insecurity (right), every city labelled. n = 17; Spearman coefficients in panel titles.

City ranks by food price premium and by deprivation (H5, descriptive)

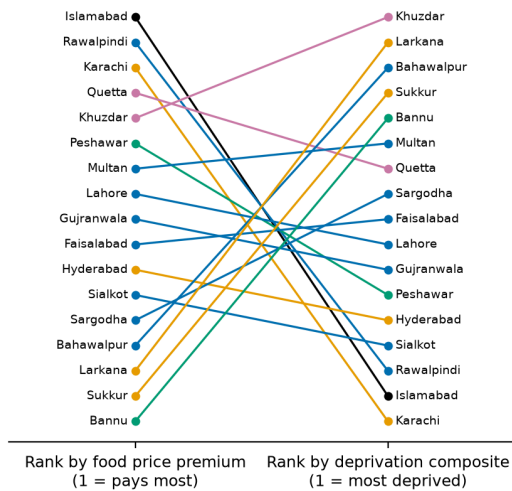


Fig. 8. Ranks by premium and by deprivation; crossing lines show the re-ranking.

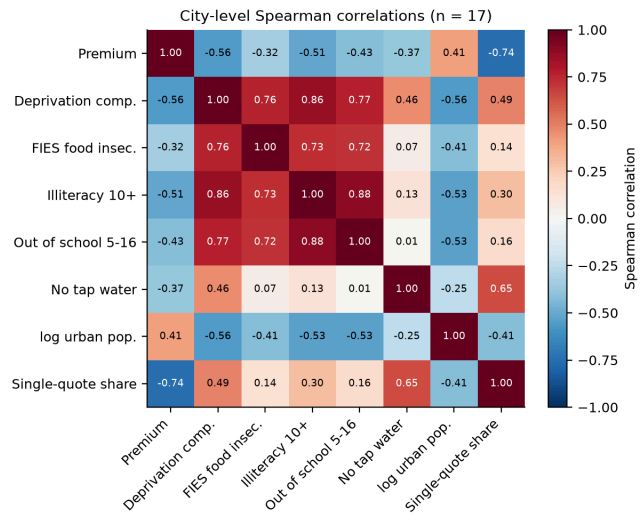


Fig. 9. City-level Spearman correlations (n = 17). Premium correlates negatively with deprivation and with the share of single-quote records.

The correlation matrix (Fig. 9) adds a warning. The premium correlates at $r = -0.68$ with the city's share of single-quote records. The high-premium cities are also those whose averages rest on more distinct quotes. Whether that reflects market structure (larger, more varied markets) or measurement is a question for H4, and it is why measurement quality is carried as a control.

3.6 Representation and measurement quality (H4, exploratory)

The 17 cities cover 65% of the urban population of the four provinces and Islamabad (Census 2023), and no rural market, AJK or Gilgit-Baltistan. Representation is uneven: Sindh holds 31% of that urban population but 24% of the cities, while KP's two cities cover only 32% of KP's urban population (Appendix Table A4). Measurement quality varies more. The share of food records where MIN = MAX, meaning the published average may rest on one quote, runs from 20% in Karachi to 91% in Sukkur (Fig. 10). Across province, population tercile and deprivation tercile, Kruskal-Wallis tests on $n = 17$ do not find group differences ($p = 0.751, 0.274$ and 0.069), but with three to five groups of three to eight cities they have little power. Aggregating to provinces would also hide a mean within-Punjab spread of 0.22 log points per item between the cheapest and dearest Punjab city (Fig. 6). That is the project's argument about the national headline in miniature.

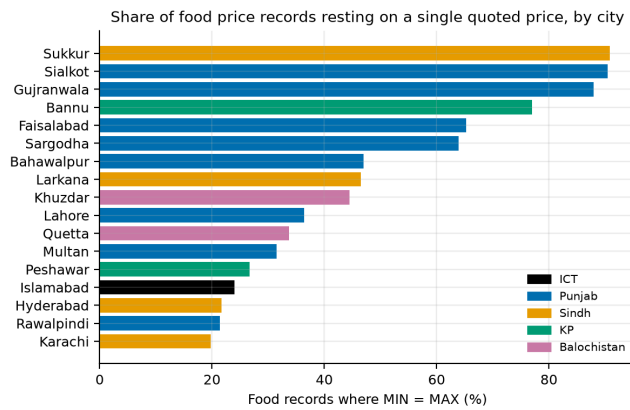


Fig. 10. Share of food records where MIN = MAX, by city: a data-quality disparity relevant to H4.

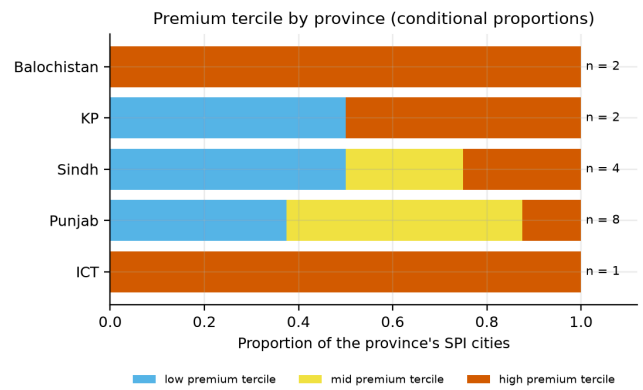


Fig. 11. Conditional proportions of premium tertiles within each province (contingency table in the notebook).

3.7 Do markets group geographically? (exploratory, later-syllabus methods)

These methods come later in the course. We use them here only to describe the data, not to confirm anything. On the standardised 17 x 31 matrix of city-by-item mean relative prices (rice IRRI-6/9 dropped for its structural gaps), Ward clustering at three clusters gives Islamabad, Rawalpindi; the other seven Punjab cities; and the Sindh, KP and Balochistan cities together (Fig. 12). Islamabad and Rawalpindi are each other's nearest neighbour, which supports treating them as one market in later robustness checks. Karachi's nearest neighbour is Hyderabad, but Hyderabad's is Faisalabad, so that pair is not mutual. PCA (Fig. 13) puts 32% of the variance on the first component, which tracks the mean premium ($|r| = 0.97$): the dominant pattern across cities is a general price level. k-means on items' dispersion, volatility and share of unchanged prices separates 2 flexible items (Onions, Tomatoes), 7 intermediate and 23 sticky items. Perishables split across all three types, so the hand-made category is a coarse proxy for price behaviour (Appendix Table A5).

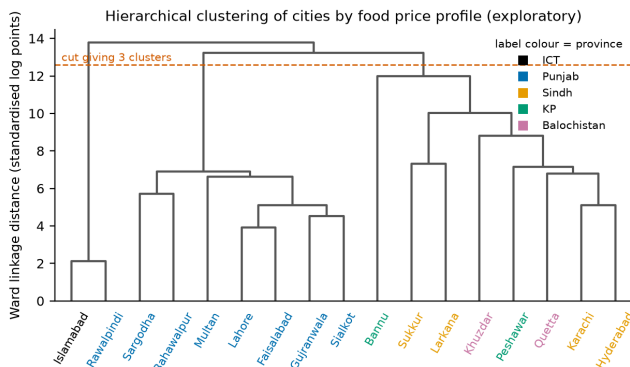


Fig. 12. Ward dendrogram of cities (exploratory); labels coloured by province.

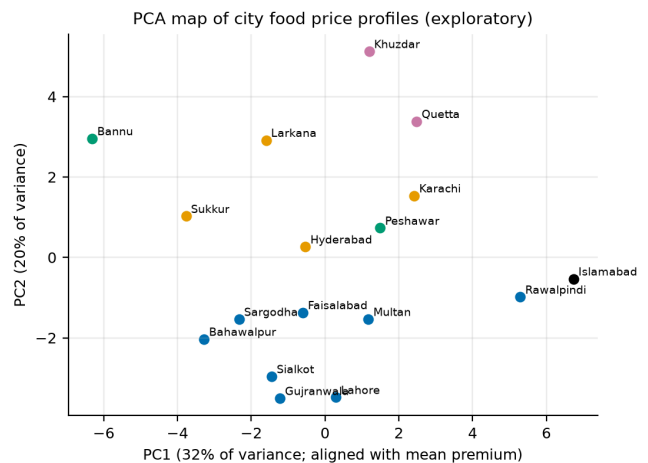


Fig. 13. Cities on the first two principal components of their food price profiles (exploratory).

4. Hypotheses and analytical plan

4.1 Formal statements

- H1a (persistent premiums; SDG 2.1):** $\log p_{c,i,t} = \alpha_{i,t} + \beta_c + \varepsilon$, with $\Sigma\beta_c = 0$. H0: $\beta_c = 0$ for all c ; H1: some $\beta_c \neq 0$. Target $\log_price$; predictors city dummies; controls item-week effects. If some cities persistently pay more for the same basket, access to food (Target 2.1) differs by place in a way the national figure hides.
- H1b (persistence):** H0: $P(\text{above median at } t+1 \mid \text{above at } t) = 0.5$; H1: > 0.5 . Consecutive pairs only, reported also for changed-price pairs, because unchanged prices make persistence partly mechanical.
- H2 (perishability and dispersion; SDG 2.c):** H0: $\mu_{CV,perishable} = \mu_{CV,storable}$; H1: perishable $>$ storable. The effective n is the number of items. The volatility half needs more consecutive weeks than the archive provides and stays contingent. Dispersion in perishables points to market-functioning frictions (Target 2.c).
- H3 (abnormal movement; SDG 2.c early warning):** H0: a classifier on lagged features does no better than the base rate ($AUC = 0.5$) on a time-ordered test; H1: $AUC > 0.5$. Target $abnormal_next_week$, with thresholds per item fitted on training weeks only.
- H4 (representation and model performance; SDG 10):** H0: error rates and data quality are equal across city groups (province, population tertile, deprivation tertile); H1: at least one group differs.
- H5 (burden versus price; SDG 2.1 and 10):** H0: Spearman $\rho(\text{premium}, \text{deprivation}) = 0$; H1: $\rho \neq 0$, two-sided, $n = 17$. The burden re-ranking is descriptive and is shown under two definitions (composite and FIES).

4.2 Refinement of H3 (sharpens the contract, does not change its substance)

Because 75% of weekly food changes are zero, the per-item 90th percentile of $|\text{change}|$ is zero for 13 food items, so "abnormal" would mean "moved at all" (base rate 7.8%). We propose defining the threshold as the 90th percentile among *non-zero* changes. That keeps "abnormal against the item's own history" but excludes 4 items with fewer than five changes, and gives a base rate of 2.9% on 7,095 labelled rows. Ranking candidate predictors with the mean-and-variance test on training weeks puts `is_perishable` first (score 6.85), followed by whether the price changed this week and the size of the change (Table 8).

Table 8. H3 candidate predictors ranked by $|\text{mean}(\text{abnormal}) - \text{mean}(\text{normal})| / \text{SE}(A - B)$, refined label, training weeks only.

feature	mean_abnormal	mean_normal	n_abnormal	n_normal	score
is_perishable	0.624	0.389	205	6890	6.854
price_changed	0.587	0.285	126	4131	6.770
abs_dlog	0.075	0.023	126	4131	4.254
log_pop_urban	14.056	14.372	205	6890	3.070
deprivation_composite	0.467	0.426	205	6890	2.884
abs_dlog_lag1	0.082	0.026	61	1831	2.680
abs_rel_price	0.105	0.086	205	6890	2.434
rel_price	-0.006	0.000	205	6890	0.614
single_quote	0.473	0.470	205	6890	0.087
within_city_range_pct	6.815	6.832	205	6890	0.021

Table 9. Hypothesis mapping: target, predictors, controls, unit of analysis, effective sample and planned test.

Hypothesis	Target	Predictor(s)	Controls	Unit	Effective n	Planned test
H1a premiums	log_price	city (sum-to-zero dummies)	item-week effects	city x item x week (food)	17 cities; 32 items x 27 weeks	F test of city effects; descriptive regression
H1b persistence	above median next week	above median this week	consecutive pairs only; changed-price subset	city x item pair	17 cities (rate per city)	one-sample t of city rates vs 0.5 (one-tailed)
H2 dispersion	mean cross-city CV	item category	none (item-level)	item	11 vs 10 items	Welch t, one-tailed; Mann-Whitney check
H3 abnormal move	abnormal_next_week	lagged change, range, single quote, perishable	item thresholds fit on training weeks	city x item x week	7,095 labelled rows; 2.9% positive	AUC on a time-ordered test (M03)
H4 representation	error rate; data quality	province, population tercile, deprivation tercile	item category	city (17) / city-group	17 cities, 3 to 5 groups	Kruskal-Wallis now; group error rates in M03
H5 burden	city premium	deprivation composite; FIES	leave-one-out	city	17	Spearman, two-sided

4.3 Validation design and planned models (Milestones 03 and 04)

Validation will be time-ordered: model fitting on the earliest weeks, validation on later ones, and the frozen week-14 holdout (weeks after 2026-09-24) opened once at the end (Fig. 14). Random folds would leak, because the same city-item series appears on both sides and later prices inform earlier labels. Accuracy alone will mislead at a 2.9% base rate: a model that always predicts "normal" is right about 97% of the time. Milestone 03 will report AUC, precision-recall and recall at a fixed alert budget, broken down by city group for H4. The planned models, within the course's agreed scope of classification, regression, clustering and fairness, are logistic regression (baseline, interpretable), tree ensembles (random forest or gradient boosting) and a small neural network for comparison, all using lag features inside the classifier. No time-series forecasting models and no optimisation.

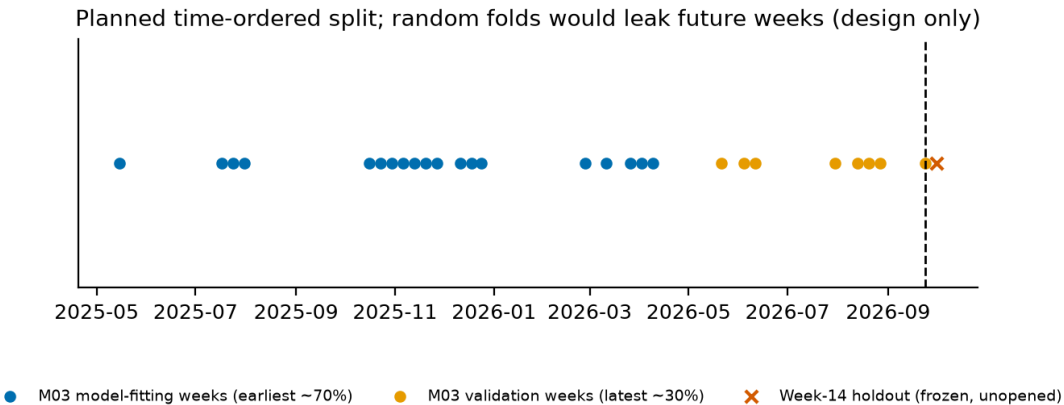


Fig. 14. Planned time-ordered split and the frozen holdout (design only; nothing was fitted in this milestone).

5. Limitations and responsible use

- **Coverage:** 27 weeks with gaps; only 15 consecutive pairs, which limits volatility and H3 sample size. More weeks will be added only from new releases, and the holdout stays sealed.
- **Purposive sample:** 17 PBS cities, urban only, with no AJK or Gilgit-Baltistan and two cities each for KP and Balochistan. No result generalises to rural or small-town markets.
- **Deprivation is not income:** district-level 2019-20 survey indicators against 2025-26 prices; a slow-moving structural attribute. District totals include rural areas, most strongly for Bannu (urban share 3.6% in Census 2023).
- **Measurement:** SPI records retail prices only. Some city averages rest on a single quote. PDF table extraction for PSLM is checked against the printed rows (page references in the manifest) and should be human-verified.
- **No causal claims:** every relationship reported here is an association across at most 17 cities.

- **Responsible use:** labelling a city "expensive" or "deprived" can stigmatise it, and a premium may reflect quality or quote mix as much as price. Results are framed as signals for further investigation, not as rankings for action.

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- Wickham, H. (2014). Tidy Data. *Journal of Statistical Software*, 59(10).
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Appendix

Table A1. Methods map: technique, course unit, where it is used, what it showed, and the CLO it serves. Rows marked (exploratory) are later-syllabus methods used in exploratory form only.

Technique	Course unit/week	Where	What it showed	CLO
Attribute types / codebook	02	\$2.4, App. C	46 columns typed; IDs nominal, rel_price interval	CLO1
Tidy data, melting	03	\$2.2	179x25 sheet with 3 stacked blocks -> 867 tidy rows/week	CLO1
Cleaning	03	\$2.2, Table 2	83 zero triples to missing; 17 relabelled rows; 0 deleted	CLO1
Missing-data options	03	\$2.3, Table 3	Three kinds of missingness; none imputed	CLO1, CLO3
Integration: stack, merge	03	\$2.5, Table 5	Row count 23,409 unchanged by every join	CLO1
Aggregation	04	\$3.6, Fig. 6	Within-Punjab range of 0.22 log pts vanishes at province level	CLO4
Splitting dates	04	\$2.4	13 calendar months; year, month, ISO week	CLO1
Sampling, bootstrap	04	\$3.1	6 city medians exclude 0 (2000 resamples)	CLO2
Feature subset selection	04	\$4, Table 8	log MIN/AVG $r = 0.998$; top H3 candidate: is_perishable	CLO2
Discretisation	04	\$2.4, Table 4	Population equal-width bins 15/0/2 vs terciles 6/5/6	CLO1
Normalisation	04	\$2.4	Min-max deprivation, z within item-week; decimal scaling rejected	CLO1
Encoding	04	\$2.4	Province/category dummies; sum-to-zero city coding	CLO1
Central tendency, geometric mean	05	\$2.1	Geometric mean reproduces PBS average in 1,377/1,377 item-weeks	CLO4
Variation, CV	05	\$3.2, Fig. 3	Mean CV perishable 0.157 vs branded 0.026	CLO4
Resistant statistics	05	\$3.1	Larkana: mean and median differ by 4.8 pts	CLO4
Outliers, both methods	05	\$3.4, Fig. 5	rel_price: 3,649 (1.5 IQR) vs 434 (3 SD)	CLO4
Histogram, bar, box plots	05	Figs. 3, 10, 5, 4	Dispersion by category; single-quote share; fences	CLO4
Scatter plots, Anscombe	05	\$3.5, Fig. 7	Leave-one-out Spearman -0.72 to -0.48	CLO4
Covariance, correlation	05	\$3.5, Fig. 9	Premium vs single-quote share $r = -0.68$	CLO4
Quant x cat, cat x cat	05	Figs. 11, 4	Premium tercile by province; rel_price by province	CLO4
Normal, CLT	06	\$3.4	rel_price excess kurtosis 3.47; t-intervals over ~32 items	CLO4
t-intervals (sigma unknown)	07	\$3.1, Fig. 2	95% CI per city, n = food items	CLO2, CLO4
Hypothesis tests, errors	07	\$3.1-3.3	4 of 17 cities survive Bonferroni; H2 $p = 0.037$	CLO2, CLO4
Hierarchical clustering (exploratory)	13-14	\$3.7, Fig. 12	3 clusters: Islamabad-Rawalpindi, other Punjab, rest	CLO2
k-means items (exploratory)	13	\$3.7	2 flexible, 7 intermediate, 23 sticky items	CLO2
Nearest neighbours (exploratory)	7	\$3.7	Islamabad and Rawalpindi mutual nearest neighbours	CLO2
PCA (exploratory)	Unit 05 (named)	\$3.7, Fig. 13	PC1 32% of variance; r with premium 0.97	CLO2
Regression (exploratory)	10	\$3.1, \$3.5	City effects = simple means ($r = 1.000$); slope on deprivation -0.127	CLO2
Holdout and CV design	8	\$4.3, Fig. 14	Time-ordered split; week-14 holdout frozen	CLO2, CLO3
Bias and fairness (exploratory)	8, 15	\$3.6, Fig. 10	MIN=MAX share 20% (Karachi) to 91% (Sukkur)	CLO3
Evaluation metrics (exploratory)	6	\$4.3	Refined abnormal base rate 2.9%	CLO2

Table A2. Per-item summary over training weeks (all 51 items): mean price (PKR per PBS unit), mean cross-city CV, share of unchanged consecutive-week prices, share of records with MIN = MAX.

PBS #	item	category	mean price (PKR)	mean CV	share unchanged wk/wk	share MIN=MAX
1	Wheat flour 20kg	Storable staple	2,213.646	0.108	0.494	0.301
2	Rice basmati broken	Storable staple	218.454	0.192	0.949	0.240
3	Rice IRR1-6/9	Storable staple	156.874	0.141	0.943	0.177
4	Bread plain	Prepared food	113.288	0.072	0.988	0.919
5	Beef with bone	Perishable	1,195.088	0.110	0.906	0.340
6	Mutton	Perishable	2,211.312	0.115	0.906	0.344
7	Chicken broiler (live)	Perishable	379.066	0.074	0.075	0.549
8	Milk fresh	Perishable	208.180	0.112	0.945	0.442
9	Curd (dahi) loose	Perishable	245.620	0.164	0.945	0.458
10	Powdered milk NIDO 390g	Branded/packaged	1,128.550	0.026	0.898	0.706
11	Eggs (farm)	Perishable	289.193	0.042	0.239	0.458
12	Mustard oil (avg quality)	Storable staple	550.066	0.122	0.894	0.423
13	Cooking oil 5L tin (brand)	Branded/packaged	2,981.018	0.015	0.875	0.935
14	Vegetable ghee 2.5kg tin (brand)	Branded/packaged	1,514.523	0.017	0.894	0.946
15	Vegetable ghee 1kg pouch (brand)	Branded/packaged	589.763	0.011	0.867	0.935
16	Bananas	Perishable	170.233	0.269	0.580	0.275
17	Pulse masoor	Storable staple	277.183	0.090	0.788	0.303
18	Pulse moong	Storable staple	382.797	0.091	0.769	0.298
19	Pulse mash	Storable staple	455.341	0.079	0.792	0.312
20	Pulse gram	Storable staple	274.550	0.103	0.678	0.333
21	Potatoes	Perishable	62.326	0.237	0.537	0.264
22	Onions	Perishable	108.545	0.184	0.259	0.301
23	Tomatoes	Perishable	138.434	0.245	0.125	0.327
24	Sugar refined	Storable staple	165.730	0.036	0.392	0.429
25	Gur	Storable staple	235.110	0.114	0.843	0.290
26	Salt powdered 800g (brand)	Branded/packaged	67.129	0.068	0.965	0.824
27	Chilli powder 200g (brand)	Branded/packaged	346.754	0.041	0.965	1.000
28	Garlic	Perishable	421.912	0.169	0.604	0.233
29	Tea Lipton 190g	Branded/packaged	450.141	0.004	0.992	0.904
30	Cooked beef (hotel plate)	Prepared food	321.057	0.168	0.984	0.355
31	Cooked daal (hotel plate)	Prepared food	172.415	0.180	0.988	0.327
32	Tea prepared (cup)	Prepared food	64.085	0.158	0.976	0.556
33	Cigarettes Capstan 20s	Other non-food	246.273	0.027	0.973	0.797
34	Long cloth 57in (brand)	Other non-food	679.400	0.163	0.996	0.523
35	Shirting	Other non-food	546.212	0.293	0.973	0.214
36	Lawn printed (brand)	Other non-food	676.896	0.156	0.988	0.458
37	Georgette	Other non-food	316.572	0.208	0.980	0.370
38	Gents sandal Bata	Other non-food	2,499.000	0.000	1.000	1.000
39	Gents sponge chappal Bata	Other non-food	628.630	0.000	1.000	1.000
40	Ladies sandal Bata	Other non-food	1,428.630	0.000	1.000	1.000
41	Electricity Q1	Administered/utility	6.086	0.000	0.733	1.000
42	Gas Q1	Administered/utility	2,522.796	0.000	0.937	1.000
43	Firewood	Other non-food	1,471.809	0.214	0.910	0.298
44	Energy saver bulb (brand)	Other non-food	413.412	0.290	0.976	0.804
45	Washing soap Sufi (brand)	Other non-food	135.581	0.046	0.988	0.832
46	Match box	Other non-food	6.297	0.117	1.000	0.824
47	Petrol super	Administered/utility	305.818	0.003	0.667	0.745
48	Hi-speed diesel	Administered/utility	326.530	0.002	0.600	0.747
49	LPG 11.67kg cylinder	Administered/utility	3,921.730	0.057	0.420	0.427
50	Telephone call charges	Administered/utility	1.790	0.000	1.000	1.000
51	Toilet soap Lifebuoy (brand)	Other non-food	117.292	0.034	0.980	0.991

Table A3a. City attributes: Census 2023 urban population and PSLM 2019-20 district deprivation indicators (higher = more deprived).

city	province	urban pop. (m)	FIES mod/sev %	illiterate 10+ %	out of school 5-16 %	no tap water %	composite (0-1)
Islamabad	ICT	1.11	7.93	15.00	10.00	74.31	0.16
Rawalpindi	Punjab	4.21	9.41	18.00	10.00	77.16	0.20
Gujranwala	Punjab	3.59	10.53	26.00	15.00	95.06	0.33
Sialkot	Punjab	1.48	9.65	21.00	11.00	93.78	0.28
Lahore	Punjab	13.00	17.96	23.00	16.00	76.71	0.34
Faisalabad	Punjab	4.39	8.67	33.00	21.00	90.11	0.35
Sargodha	Punjab	1.61	16.38	33.00	18.00	92.44	0.43
Multan	Punjab	2.50	16.88	39.00	30.00	89.09	0.51
Bahawalpur	Punjab	1.62	21.25	51.00	32.00	86.87	0.61
Karachi	Sindh	18.87	9.60	23.85	25.74	32.94	0.14
Hyderabad	Sindh	2.02	16.59	34.00	34.00	31.68	0.30
Sukkur	Sindh	0.81	15.53	46.00	48.00	81.74	0.59
Larkana	Sindh	0.80	17.28	44.00	56.00	100.00	0.70
Peshawar	KP	1.91	11.22	40.00	25.00	55.51	0.30
Bannu	KP	0.05	18.73	47.00	39.00	75.19	0.56
Quetta	Balochistan	1.57	23.47	36.00	35.00	59.72	0.49
Khuzdar	Balochistan	0.36	29.26	78.00	60.00	92.98	0.97

Table A3b. City-district crosswalk and alignment decisions.

city	city_code	province	pslm_districts	census_units	pslm_rule	notes
Islamabad	01	ICT	Islamabad	ISLAMABAD	single district	Islamabad and Rawalpindi are one contiguous market but two administrative units; kept separate, each with its own district values.
Rawalpindi	02	Punjab	Rawalpindi	RAWALPINDI	single district	See Islamabad.
Gujranwala	03	Punjab	Gujranwala	GUJRANWALA	single district	–
Sialkot	04	Punjab	Sialkot	SIALKOT	single district	–
Lahore	05	Punjab	Lahore	LAHORE	single district	District is 100% urban in Census 2023.
Faisalabad	06	Punjab	Faisalabad	FAISALABAD	single district	–
Sargodha	07	Punjab	Sargodha	SARGODHA	single district	–
Multan	08	Punjab	Multan	MULTAN	single district	–
Bahawalpur	09	Punjab	Bahawalpur	BAHAWALPUR	single district	–
Karachi	10	Sindh	Karachi Central; Karachi East; Karachi Malir; Karachi South; Karachi West; Korangi	KARACHI CENTRAL; KARACHI EAST; MALIR; KARACHI SOUTH; KARACHI WEST; KEAMARI; KORANGI	population-weighted mean of 6 PSLM districts (Census 2023 weights; Keamari added to Karachi West)	Seven census districts form one SPI market; PSLM 2019-20 reports six (Keamari created later).
Hyderabad	11	Sindh	Hyderabad	HYDERABAD	single district	–
Sukkur	12	Sindh	Sukkur	SUKKUR	single district	–
Larkana	13	Sindh	Larkana	LARKANA	single district	–
Peshawar	14	KP	Peshawar	PESHAWAR	single district	–
Bannu	15	KP	Bannu	BANNU	single district	Census 2023 urban share of Bannu district is 3.6%; district totals describe a mostly rural district.
Quetta	16	Balochistan	Quetta	QUETTA	single district	–
Khuzdar	17	Balochistan	Khuzdar	KHUZDAR	single district	PSLM 2019-20 has no urban sample for Khuzdar: urban indicators are missing, district totals used.

Table A4. Representation: SPI cities' urban population against provincial urban population (Census 2023).

province	n_cities	spl_pop_urban	province_pop_urban	share_of_province_urban	share_of_national_urban	share_of_spl_cities
ICT	1	1,108,872.000	1,108,872.000	1.000	0.012	0.059
Punjab	8	32,412,617.000	51,975,967.000	0.624	0.560	0.471
Sindh	4	22,503,550.000	28,965,488.000	0.777	0.312	0.235
KP	2	1,954,373.000	6,131,296.000	0.319	0.066	0.118
Balochistan	2	1,929,924.000	4,611,828.000	0.418	0.050	0.118

Table A5. k-means market types (exploratory) against hand-made item categories (number of food items).

item_category	flexible	intermediate	sticky
branded_packaged	0	0	7
perishable	2	5	4
prepared_food	0	0	4
storable_staple	0	2	8

Table A6. H5 correlations under three deprivation definitions, with leave-one-out ranges (n = 17).

deprivation_measure	n	pearson_r	pearson_p	spearman_rho	spearman_p	loo_pearson_min	loo_pearson_max	loo_spearman_min	loo_spearman_max	loo_most_influential
deprivation_composite	17	-0.494	0.044	-0.559	0.020	-0.730	-0.397	-0.721	-0.476	Khuzdar
fies_mod_sev_pct	17	-0.335	0.188	-0.321	0.209	-0.491	-0.185	-0.456	-0.185	Khuzdar
deprivation_composite_nowater	17	-0.443	0.075	-0.500	0.041	-0.664	-0.308	-0.659	-0.400	Khuzdar

Table A7. Outlier counts by the quartile and mean/SD methods, both tiers.

variable	n	Q1	Q3	IQR	mean	sd	iqr_mild_1.5	iqr_regular_3	sd_mild_2	sd_regular_3
rel_price, all 51 items	23326	-0.0374	0.0464	0.0838	0.0000	0.1328	3649	1251	1598	434
weekly log change, all items	12959	0.0000	0.0000	0.0000	0.0008	0.0620	2502	2502	493	296

Table A8. Codebook of the master dataset (Unit 02 attribute types; missing counts).

column	attribute_type	meaningful_operations	n_missing	pct_missing
week_end	interval	differences in days; order; no true zero	0	0.00
year	interval	order, differences	0	0.00
month	ordinal (cyclic)	order within year; mode	0	0.00
iso_week	ordinal (cyclic)	order within year	0	0.00
city	nominal	equality, counts, mode	0	0.00
city_code	nominal	equality only (stored as text)	0	0.00
province	nominal	equality, counts	0	0.00
item_id	nominal	equality only (number is a PBS serial)	0	0.00
item_label	nominal	equality	0	0.00
item_short	nominal	equality	0	0.00
unit	nominal	equality	0	0.00
is_food	binary (nominal)	counts, proportions	0	0.00
item_category	nominal	counts, mode	0	0.00
price_min	ratio	all arithmetic; ratios meaningful within an item	83	0.35
price_avg	ratio	all arithmetic; ratios meaningful within an item	83	0.35
price_max	ratio	all arithmetic; ratios meaningful within an item	83	0.35
is_structural_missing	binary	counts	0	0.00
log_price	interval	differences = log ratios	83	0.35
national_gm	ratio	all arithmetic within an item-week	0	0.00
rel_price	interval	differences, means; ratios of rel_price are meaningless	83	0.35
cross_city_median	ratio	within an item-week	0	0.00
above_median	binary	proportions	83	0.35
above_median_prev	binary	proportions	10450	44.64
within_city_range_pct	ratio	all arithmetic	83	0.35
single_quote	binary	proportions	83	0.35
price_z_item_week	interval	differences, means	2903	12.40
days_since_prev	ratio	all arithmetic	867	3.70
is_consecutive	binary	counts	0	0.00
dlog_price	interval	differences, means	10450	44.64
price_changed	binary	proportions	10450	44.64
dlog_price_lag1	interval	differences, means	15634	66.79
pop_total	ratio	all arithmetic	0	0.00
pop_urban	ratio	all arithmetic	0	0.00
log_pop_urban	interval	differences	0	0.00
fies_mod_sev_pct	ratio	all arithmetic	0	0.00
illiteracy10_pct	ratio	all arithmetic	0	0.00
out_of_school_pct	ratio	all arithmetic	0	0.00
no_tap_water_pct	ratio	all arithmetic	0	0.00
deprivation_composite	interval (0-1 scaled)	order, differences; zero is the sample minimum	0	0.00
deprivation_composite_nowater	interval (0-1 scaled)	order, differences	0	0.00
cpi_urban_food	ratio (index)	ratios between months	0	0.00
real_price_avg	ratio	all arithmetic	83	0.35
block	nominal	provenance only	0	0.00
source_file	nominal	provenance only	0	0.00
sha256	nominal	provenance only	0	0.00
parser_version	nominal	provenance only	0	0.00